

FIT5225 Assignment 2

Individual Report

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Repository: <https://github.com/chenjunboa/PacificBioArchive>

Role and Contribution

My role was to complete the final user-facing integration, stabilize the handed-off backend paths needed by the interface, and prepare the project for demonstration and final acceptance. I completed the Cognito-aware React workflow for sign-up, sign-in, session restoration, token refresh, and sign-out, while keeping a local development login path for automated testing. I implemented the upload, query, media management, and notification UI flows, including file type and size feedback, video duration validation, checksum progress, polling, duplicate upload messaging, four query modes, bulk tag add/remove, owner-only delete, and SNS subscription messaging.

During handoff review I found that several cloud-mode backend paths were not ready for the final UI to demonstrate truthfully. I helped stabilize DynamoDB tag-count indexes, thumbnail reverse mappings, temporary query reservations, delete cleanup, and worker index writes. I also added and ran Playwright end-to-end tests, prepared the demo script and evidence checklist, deployed the latest API/Worker Lambda images, disabled the fallback worker that was consuming messages, and validated the live cloud path with two image smoke runs and one short MP4 smoke run.

Reflection

This project showed me how quickly a group cloud assignment can become an integration problem rather than four isolated coding tasks. Earlier members established important foundations, especially the local prototype and AWS/GCP deployment, but the later handoff exposed gaps around query indexes, cleanup consistency, real worker routing, and evidence boundaries. I learned that it is not enough for code to exist in the repository; we had to prove that the deployed system was using the expected version, that the correct worker consumed SQS messages, and that records were cleaned up after testing.

The team workflow improved when we treated unresolved items as explicit handoff risks instead of quiet assumptions. My biggest challenge was balancing urgency with honesty: I wanted the report to sound complete, but only after real validation. The final image and video smoke tests gave us a much stronger basis for the report and demo. If I repeated the project, I would ask the team to define shared evidence requirements

earlier, including who owns cloud screenshots, email confirmations, and final release approval.

Generative AI Use

I used OpenAI Codex selectively to inspect the repository, compare the implementation with the assignment brief, generate focused code changes, run tests, debug AWS/GCP integration issues, and draft documentation. I reviewed the generated changes through builds, Playwright tests, AWS CLI checks, CloudWatch logs, DynamoDB cleanup checks, and Git history before accepting them. No cloud secrets, Cognito passwords, verification codes, JWTs, presigned URL query strings, Terraform state, or model weights were committed.